

Assignment: CCNA - VLANs and Inter - VLAN Routing

1- Which of the following messages in the DHCP process are broadcasted? (Choose two)

- A. Request
- B. Offer
- C. Discover
- D. Acknowledge

Answer:-

- B. Offer**
- C. Discover**

2- Which command would you use to ensure that an ACL does not block web-based TCP traffic?

- A. permit any
- B. permit tcp any any eq 80
- C. permit tcp any eq 80
- D. permit any any eq tcp

Answer:-

- B. permit tcp any any eq 80**

3- R1, R2, R3, and R4 have their Fast Ethernet 0/0 interfaces attached to the same VLAN. A network engineer has typed a configuration for each router by using a word processor. He will later copy and paste the configuration into the routers. Examine the following exhibit, which lists configuration for the four routers, as typed by the network engineer. Assuming that all four routers can ping each other's LAN IP addresses after the configuration has been applied, choose the routers that will be able to form a neighbor relationship with the other routers on the LAN. (You can assume that, if not shown in the exhibit, all other related parameters are still set to their defaults.) (Choose two)

```
Configuration (
!  
interface faste  
    ip address :  
!  
router ospf 0  
    network 10.0
```

```
Configuration (
!  
interface faste  
    ip address :  
!  
router ospf 1  
    network 10.0
```

```
Configuration (
!  
interface faste  
    ip address :  
!  
router ospf 3  
    network 10.0
```

```
Configuration (
!  
interface faste  
    ip address :  
!  
router ospf 2  
    network 10.0
```

- A. R1
- B. R2
- C. R3
- D. R4
- E. None of the routers will exchange routing information.

Answer:-

- B. R2
- D. R4

4-enable secret [password] is hashed using the _____ algorithm.

- A. MD5
- B. AH

- C. PSK
- D. ESP
- E. WPA2

Answer:-

A. MD5

5- An engineer connects to Router R1 and issues a show ip ospf neighbor command. The status of neighbor 2.2.2.2 lists FULL/BDR. What does the BDR mean?

- A. R1 is an Area Border Router.
- B. R1 is a backup designated router.
- C. Router 2.2.2.2 is an Area Border Router.
- D. Router 2.2.2.2 is a backup designated router.

Answer:-

D. Router 2.2.2.2 is a backup designated router.

6- Which command is used to view the neighbor discovery table on a PC?

- A. show ipv6 neighbor
- B. show ipv6 neighbors
- C. netsh interface ipv6 show neighbor
- D. netsh interface ipv6 show neighbors

Answer:-

C. netsh interface ipv6 show neighbor

7- What type of variable is being shown? Routers = [R1,R2,R3]

- A. List
- B. Dictionary
- C. Simple
- D. Unsigned integers

Answer:-

A. List

8- Identify the fields in an IPv4 header. (Choose three)

- A. Host component
- B. Time to Live
- C. Source address
- D. Destination address
- E. Network address

Answer:-

B. Time to Live

C. Source address

D. Destination address

9- Host A and Host B sit in two different subnets. The path between the subnets of these two hosts runs through three different Layer 3 forwarding devices (routers and Layer 3 switches). A network engineer uses the APIC-EM Path Trace ACL Analysis tool to analyze the path used for Host A to send packets to Host B. Which part of the function is done specifically by the ACL Analysis or ACL Trace part of the tool?

- A. Discovery of the topology that exists between the two hosts
- B. Analysis of the Layer 3 forwarding decisions in the path from Host A to B
- C. Analysis of the Layer 2 forwarding decisions in the path from Host A to B
- D. Analysis of the impact of ACLs on the packets that would flow from Host A to B

Answer:-

D. Analysis of the impact of ACLs on the packets that would flow from Host A to B

10- Which IPv6 address is the equivalent of the IPv4 interface loopback address 127.0.0.1?

- A. ::1
- B. ::
- C. 2000::/3
- D. 0::/10

Answer:-

A. ::1

11- Which command is used to apply an ACL to an interface?

- A. access-group
- B. ip access-group
- C. ip access-list
- D. ip access-class
- E. access-class
- F. access-list

Answer:-

B. ip access-group

12- Which command and mode will successfully configure a hostname of R1 on a Cisco IOS router?

- A. Router(config)#name R1

B. Router# hostname R1

C. Router(config)#hostname R1

D. Router#name R1

E.Router>hostname R1

F. Router>name R1

Answer:-

C. Router(config)#hostname R1

13- Which of the following reserved IPv4 addresses has binary 0s in all of the host bit positions?

A. Local broadcast address

B. Loopback address

C. Directed broadcast address

D. Network address

E. All zeros address

Answer:-

D. Network address

14- A Cisco Catalyst switch connects to what should be individual user PCs. Each port has the same port security configuration, configured as follows:

interface range gigabitethernet 0/1- 24

switchport mode access

switchport port-security

switchport port-security mac-address sticky

Which of the following answers describe the result of the port security configuration created with these commands? (Choose two)

A. Prevents unknown devices with unknown MAC addresses from sending data through the switch ports.

B. If a user connects a switch to the cable, prevents multiple devices from sending data through the port.

C. Will allow any one device to connect to each port, and will save that device's MAC address into the startup-config

D. Will allow any one device to connect to each port, but will not save that device's MACaddress into the startup-config

Answer:-

B. If a user connects a switch to the cable, prevents multiple devices from sending data through the port.

C. Will allow any one device to connect to each port, and will save that device's MAC address into the startup-config

15- What is the Administrative Distance of internal EIGRP routes?

B. 170

C. 90

D. 20

E. 1

F. 110

G. 120

Answer:-

C. 90

16- When a subnet mask is presented in binary, what do the binary 1s represent?

A. The network portion of an associated address

B. The host portion of the subnet mask

C. The number of wildcard bits in the subnet mask

D. The number of wildcard bits in the address

E. The network portion of the subnet mask

F. The host portion of an associated address

Answer:-

A. The network portion of an associated address

17- Which switch would STP choose to become the root bridge in the selection process?

A. 32768: 11-22-33-44-55-66

B. 32768: 22-33-44-55-66-77

C. 32769: 11-22-33-44-55-65

D. 32769: 22-33-44-55-66-78

Answer:-

A. 32768: 11-22-33-44-55-66

18- Which of the following devices is used by the service provider to provide WAN services?

- A. Router**
- B. Core router**
- C. WANswitch**
- D. CSU/DSU**

Answer:-

D. CSU/DSU

19- Your Cisco IOS router is acting as a DHCP server.

Which command will display the addresses that have been handed out to clients on the LAN?

- A. show ip dhcp assignments**
- B. show ip dhcp address**
- C. show ip dhcp conflicts**
- D. show ip dhcp bindings**
- E. show ip dhcp pool**

Answer:-

D. show ip dhcp bindings

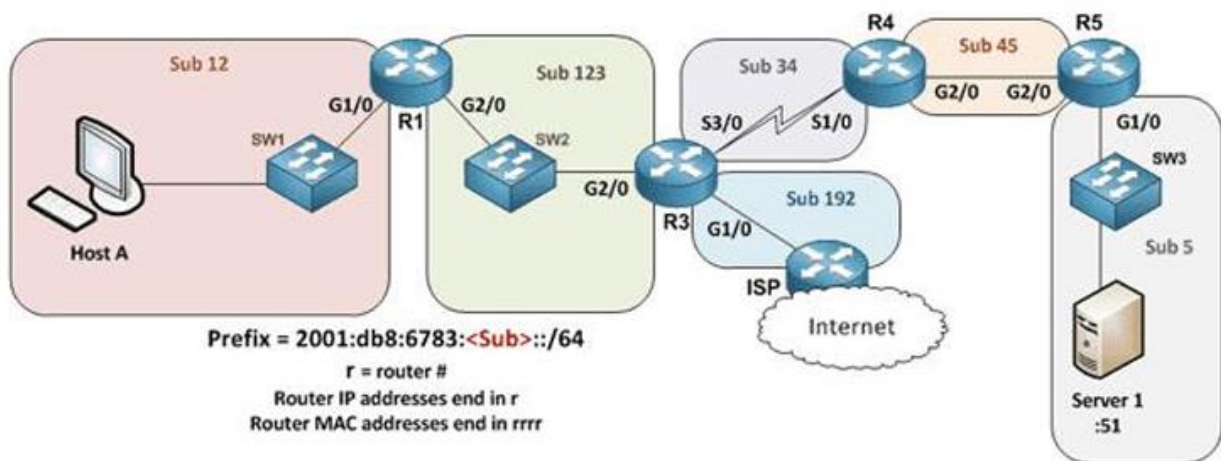
20- Which of the following commands would you use to enable EIGRP only on those interfaces with an IP address from 10.1.1.0 through 10.1.1.63?

- A. network 10.1.1.0 0.0.0.63**
- B. network 10.1.1.0/63**
- C. router eigrp 10.1.1.0 0.0.0.63**
- D. network 10.0.0.0 0.0.0.255**

Answer:-

A. network 10.1.1.0 0.0.0.63

21- R3 has a static route configured that points toward the service provider. What command could you use to have R3 advertise an OSPFv3 default route to the internal network, regardless of whether R3 had its default static route?



- A. The decision to advertise a default route depends on the static route always being present on R3.
- B. The default behavior is to redistribute any default IPv6 routes into OSPFv3, so no action is required.
- C. Each of the other routers needs a static default route that leads to R3.
- D. Use the command `default-information originate` always in interface mode for G1/0 on R3.
- E. Have R3 use the command `default-information originate` always in OSPFv3 router configuration mode.

Answer:-

E. Have R3 use the command `default-information originate` always in OSPFv3 router configuration mode.

22- You are configuring dynamic NAT on your Cisco IOS router. Which command is used to verify the interfaces that are being used as the outside interface and the inside interface?

- A. `show interfaces`
- B. `show ip route`
- C. `show ip nat translations`
- D. `show ip interface brief`
- E. `show ip interface`
- F. `show ip nat statistics`

Answer:-

E. `show ip interface`

23- When using the “`show EtherChannel summary`” command, what does the “u” flag signify?

- A. Waiting to be aggregated

- B. Suspended**
- C. In use**
- D. Unsuitable for bundling**

Answer:-

C. In use

24- Which command could you enter to encrypt passwords?

- A. enable secret**
- B. username {username} secret {password}**
- C. service password-encryption**
- D. All of the above**
- E. None of the above**

Answer:-

D. All of the above

25- You are setting up a Cisco IOS router as a DHCP server.

Which command is used to identify the IPv4 addresses that will be in the DHCP pool?

- A. network**
- B. dns-server**
- C. default-router**
- D. ip dhcp excluded-address**
- E. lease**
- F. ip dhcp pool**
- G. domain name**

Answer:-

A. network

26- Which of the following statements are true regarding the processing of ACLs that have been applied to router interfaces? (Choose two)

- A. Inbound ACLs will be processed before the routing table lookup occurs**
- B. Inbound ACLs will be processed after the routing table lookup has occurred**
- C. Outbound ACLs will be processed after the routing table lookup has occurred**
- D. Outbound ACLs will be processed before the routing table lookup occurs**

Answer:-

A. Inbound ACLs will be processed before the routing table lookup occurs

C. Outbound ACLs will be processed after the routing table lookup has occurred

27- imagine you configured OSPFv2 in a small lab network. Which of the following answers list a condition that could keep the routers in your lab from learning all the routes to all the IPv4 routes in your small lab network? (Choose two)

- A. An ACL could be blocking router advertisements.**
- B. Two neighbouring routers that connect to the same link have been configured with the same OSPF area and with the same IPv4 subnet mask.**
- C. Any physical layer problem that would prevent two neighbouring routers from being able to ping each others IPv4 addresses in the subnet that exists between the two routers.**
- D. Two neighboring routers that connect to the same link have been configured with the same OSPF process ID on the router ospf command.**

Answer:-

- A. An ACL could be blocking router advertisements.**
- C. Any physical layer problem that would prevent two neighbouring routers from being able to ping each others IPv4 addresses in the subnet that exists between the two routers.**

28- Which statements describe neighbor discovery functionality in IPv6? (Choose two)

- A. Determines the link layer address of a neighbor**
- B. Finds neighbor switches on the link**
- C. Is achieved by using Dynamic Host Configuration Protocol for IPv6, or DHCPv6 with IPv6 multicast**
- D. Queries for duplicate addresses**

Answer:-

- A. Determines the link layer address of a neighbor**
- D. Queries for duplicate addresses**

29- Which IPv6 prefix will the typical enterprise network receive from the service provider?

- A. /52**
- B. /56**
- C. /64**
- D. /32**
- E. /48**
- F. /60**

Answer:-

E. /48

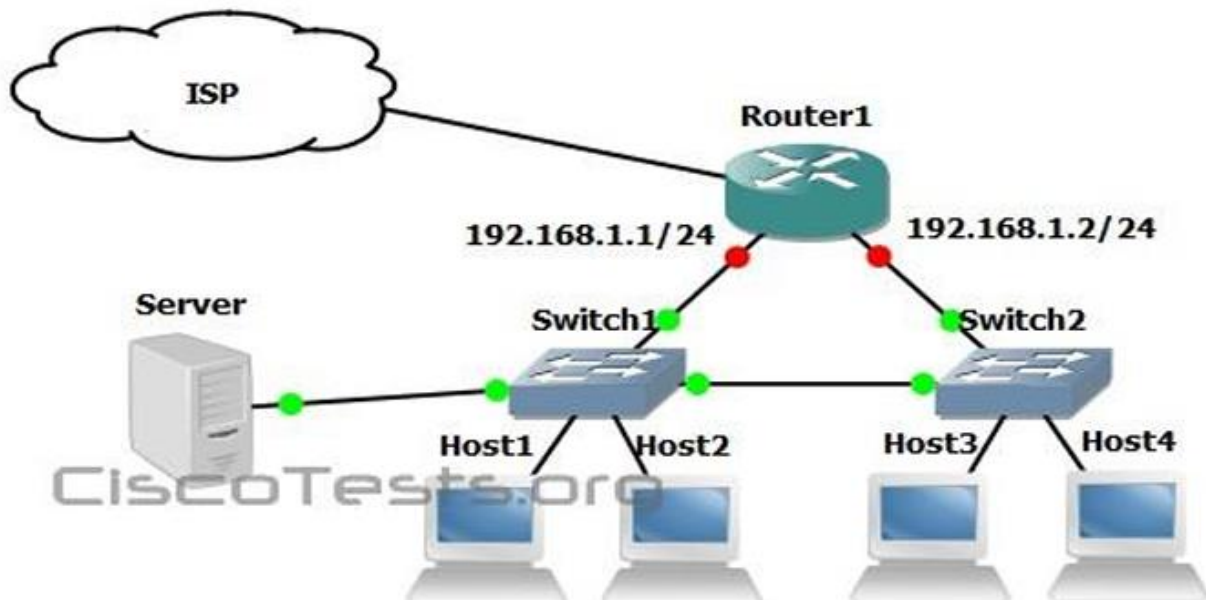
30- How should be configured a switch so that it could be accessed remotely?

- A. Assign a password and privilege level**
- B. Apply the access control list, or ACL, to the virtual type terminal, or vty, lines**
- C. Configure a gateway for the switch**
- D. Generate a certificate**

Answer:-

C. Configure a gateway for the switch

31- Refer to the exhibit. A network technician is asked to design a small network with redundancy. The exhibit represents this design, with all hosts configured in the same VLAN. What conclusions can be made about this design?



- A. This design will function as intended.**
- B. Spanning-tree will need to be used.**
- C. The router will not accept the addressing scheme.**
- D. The connection between switches should be a trunk.**
- E. The router interfaces must be encapsulated with the 802.1Q protocol.**

Answer:-

B. Spanning-tree will need to be used.

D. The connection between switches should be a trunk.